UNIT
10

SURFACE CHEMISTRY



Irving Langmuir

Irving Langmuir was an American Chemist and Physicist. He was awarded Nobel Prize in the year 1932 in Chemistry for his works in Surface Chemistry. He outlined the concentric theory of atomic structure. He invented the gas filled incandescent lamp and hydrogen welding technique. The Langmuir Laboratory for Atmospheric Research near Socorro, New Mexico was named in his honor. Langmuir and Tonks discovered electron density waves in plasmas that are now known as Langmuir waves.

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Learning Objectives

After studying this unit the student will be able to

- classify adsorption.
- distinguish between absorption and adsorption
- explain Freundlich adsorption isotherm
- understand catalysis and the characteristics of catalysts.
- explain the theories of catalysis and enzyme catalysis.
- classify colloids.
- explain the methods of preparation and purification of colloids.
- discuss the properties of colloidal solution.
- explain the role of colloids and emulsions in daily life.

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